01617 Introduction to Dynamical Systems (E25) - Assignment 2

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1) Problem 1 — The Duffing Equation

We are given another flavour of the Duffing equation:

$$\ddot{x}_1 = f(x_1, x_2) = \mu x_1 - x_1^3 \tag{1}$$

or equivalently as a system of first order ODEs:

$$\begin{aligned} \dot{x}_1 &= f_1(x_1, x_2) = x_2 \\ \dot{x}_2 &= f_2(x_1, x_2) = \mu x_1 - x_1^3 \end{aligned} \tag{2}$$

with $\mu \in \mathbb{R}$ and $(x_1, x_2) \in E = \mathbb{R}^2$. We use the suppressed notation for $x_{1,2}(t) : \mathbb{R}^+ \to \mathbb{R}$ such that $\dot{x}_{1,2} = \frac{\mathrm{d}x_{1,2}}{\mathrm{d}t}$, with higher order derivatives being indicated by additional dots.

1.a) Finding critical points

We wish to find values of x_1, x_2 such that $f_1(x_1, x_2) = \dot{x}_1 = 0$ and $f_2(x_1, x_2) = \dot{x}_2 = 0$. Thus, we find that:

$$\begin{aligned} \dot{x}_1 = 0 \quad &\Rightarrow \quad x_2 = 0 \\ \dot{x}_2 = x_1(\mu - x_1^2) = 0 \quad &\Rightarrow \quad x_1 \in \{0, +\sqrt{\mu}, -\sqrt{\mu}\} \end{aligned} \tag{3}$$

From this, we can conclude that the critical points of the system are given by:

$$\mathbf{x}^* \in \begin{cases} \{(0,0)\} & \forall \mu \in \mathbb{R} \\ \{(-\sqrt{\mu}, 0), (0,0), (\sqrt{\mu}, 0)\} & \forall \mu > 0 \\ \{(0,0), \cancel{(-i\sqrt{|\mu|}, 0)}, \cancel{(i\sqrt{|\mu|}, 0)}\} & \forall \mu < 0 \end{cases} \tag{4}$$

Where the $\mu < 0$ case produces two complex critical points which are discarded due to the domain restriction $(x_1, x_2) \in \mathbb{R}^2$. Hence we will investigate the system behaviour around the critical points

$$\mathbf{x}^* \in \begin{cases} \{(-\sqrt{\mu}, 0), (0,0), (\sqrt{\mu}, 0)\} & \mu > 0 \\ \{(0,0)\} & \mu \leq 0 \end{cases} \tag{5}$$

for easier reference later, we enumerate the critical points as follows:

$$\mathbf{x_0^*} = (0,0) \quad \mathbf{x_1^*} = (\sqrt{\mu}, 0) \quad \mathbf{x_2^*} = (-\sqrt{\mu}, 0) \tag{6}$$

1.b) Stability of critical points

We will analyse the stability of the critical points found in Equation 5 by computing the Jacobian matrix of the system at each critical point and evaluating its eigenvalues. Here we will exclude the case for $\mu = 0$.

The Jacobian matrix $A = D\mathbf{f}(\mathbf{x})$ of the system/vector field $\mathbf{f} = (f_1, f_2)^\mathsf{T} \in C^1(E)$ from Equation 2 given by:

$$D\mathbf{f}(x_1, x_2) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \mu - 3x_1^2 & 0 \end{bmatrix} \tag{7}$$

1.b.1) Case: $\mu < 0$

We consider the case $\mu < 0$ first, which only has a single critical point at the origin $\mathbf{x_0^*} = (0,0)$, at which we linearise the system by finding the Jacobian matrix:

$$A = D\mathbf{f}(\mathbf{x_0^*}) = \begin{bmatrix} 0 & 1 \\ \mu & 0 \end{bmatrix} \tag{8}$$

Which we obtain the eigenvalues λ_i of by solving the characteristic polynomial:

$$p_{x_0^*}(\lambda) = \det(A - \lambda I) = \lambda^2 - \mu = 0 \Rightarrow \lambda_{1,2} = \pm\sqrt{\mu} = \pm i\sqrt{|\mu|} \tag{9}$$

Which form a pure imaginary complex conjugate pair, indicating that the critical point is a *center* for the linearised system. We seek to employ [1, Chapter 2.10, Theorems 5 & 6], which requires that we investigate the symmetry of the vector field $\mathbf{f}$. Using the definition of symmetry from [1, Chapter 2.10, Definition 6], we consider a transformation $(t, x_1) \to (-t, -x_1)$ applied to Equation 1:

$$\ddot{x}_1 = \mu x_1 - x_1^3 \Leftrightarrow -\ddot{x}_1 = -\mu x_1 + x_1^3 \tag{10}$$

Which indicates that the vector field is symmetric with respect to the x_2 axis. Thus, by [1, Chapter 2.10, Theorem 6], we conclude that the critical point $\mathbf{x_0^*} = (0, 0)$ is also a *center* of the non-linear system.

In order to gauge the stability of the critical point, we employ the stability theory presented in [1, Chapter 2.9], where Definition 1 defines stability as:

Definition 1: Let ϕ_t denote the flow of the differential equation $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ defined for all $t \in \mathbb{R}$. An equilibrium point $\mathbf{x}_0$ is *stable* if for all $\varepsilon > 0$ there exists a $\delta > 0$ such that for all $\mathbf{x} \in N_\delta(\mathbf{x}_0)$ we have

$$\phi_t(\mathbf{x}) \in N_\varepsilon(\mathbf{x}_0) \tag{11}$$

The equilibrium point $\mathbf{x}_0$ is *unstable* if it is not stable. And $\mathbf{x}_0$ is *asymptotically stable* if it is stable and if there exists a $\delta > 0$ such that for all $\mathbf{x} \in N_\delta(\mathbf{x}_0)$ we have

$$\lim_{t\to\infty} \phi_t(\mathbf{x}) = \mathbf{x_0} \tag{12}$$

We start by constructing a Liapunov function $V(\mathbf{x})$ for the system, which by [1, Chapter 2.9, Theorem 3] must satisfy:

$$V(\mathbf{x_0}) = 0 \qquad \wedge \qquad V(\mathbf{x}) > 0 \text{ if } \mathbf{x} \neq \mathbf{x}_0 \tag{13}$$

We construct such a function by considering energy conservation of the system and production of the Hamiltonian, H, which by definition satisfies:

$$\frac{\partial H}{\partial x_2} = \dot{x_1} = x_2 \qquad \frac{\partial H}{\partial x_1} = \dot{x_2} = -\mu x_1 + x_1^3 \tag{14}$$

Such that:

$$\begin{aligned} H(x_1, x_2) &= \int x_2 \, \mathrm{d}x_2 = \frac{1}{2}x_2^2 + h_1(x_1) \\ H(x_1, x_2) &= \int -\mu x_1 + x_1^3 \, \mathrm{d}x_1 = -\frac{1}{2}\mu x_1^2 + \frac{1}{4}x_1^4 + h_2(x_2) \end{aligned} \tag{15}$$

Allowing the construction of the Hamiltonian, which we let be our Liapunov function:

$$V(\mathbf{x}) = H(x_1, x_2) = \frac{1}{2}x_2^2 - \frac{1}{2}\mu x_1^2 + \frac{1}{4}x_1^4 + \varkappa \tag{16}$$

Which satisfies the necessary conditions for a Liapunov function given in Equation 13 by inspection. We consider the temporal derivative of $V(\mathbf{x})$:

$$\dot{V}(\mathbf{x}) = 0 \tag{17}$$

Which by [1, Chapter 2.9, Theorem 3] implies that the critical point is *stable*, but not *asymptotically stable*. This is sometimes referred to as *Liapunov stability* or *neutral stability*.

1.b.2) Case: $\mu > 0$

We then consider the case $\mu > 0$, which has three critical points at $\mathbf{x_0^*} = (0, 0)$, $\mathbf{x_1^*} = (\sqrt{\mu}, 0)$, and $\mathbf{x_2^*} = (-\sqrt{\mu}, 0)$.

We first compute the Jacobian at $\mathbf{x_0^*} = (0,0)$ and the eigenvalues of the linearised system:

$$A = D\mathbf{f}(\mathbf{x_0^*}) = \begin{bmatrix} 0 & 1 \\ \mu & 0 \end{bmatrix} \tag{18}$$

$$p_{x_0^*}(\lambda) = \det(A - \lambda I) = \lambda^2 - \mu = 0 \Rightarrow \lambda_{1,2} = \pm\sqrt{\mu} \tag{19}$$

Where the eigenvalues are real and of opposite signs, indicating the critical point is a *saddle point* for the linearised system. Additionally, we note that saddle points are hyperbolic as their eigenvalues are real and non-zero, and that they are *unstable* by [1, Chapter 2.9, Theorem 2].

As such, we are able to leverage [1, Chapter 2.10, Theorem 3] to conclude that the critical point $\mathbf{x_0^*} = (0,0)$ is also a *saddle point* of the non-linear system.

We may now consider the other two critical points $\mathbf{x_{1,2}^*} = (\pm\sqrt{\mu}, 0)$, for which we compute the Jacobian and eigenvalues of the linearised system:

$$A = D\mathbf{f}\left(\mathbf{x_{1,2}^*}\right) = \begin{bmatrix} 0 & 1 \\ \mu - 3x_1^2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \mu - 3\mu & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2\mu & 0 \end{bmatrix} \tag{20}$$

$$p_{x_{1,2}^*}(\lambda) = \det(A - \lambda I) = \lambda^2 + 2\mu = 0 \Rightarrow \lambda_{1,2} = \pm i\sqrt{2\mu} \tag{21}$$

By the same logic as in Section 1.b.1, we find that these are *centers* of the linearised system.

In order to determine their stability, seek to prove that they are local minimisers by computing the Hessian:

$$D^2V(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 V}{\partial x_1^2} & \frac{\partial^2 V}{\partial x_1 \partial x_2} \\ \frac{\partial^2 V}{\partial x_2 \partial x_1} & \frac{\partial^2 V}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} -1\mu + 3\mu & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2\mu & 0 \\ 0 & 1 \end{bmatrix} \tag{22}$$

Which is positive definite for $\mu > 0$, implying that these critical points are local minimisers of the Liapunov function from Equation 16. Thus, by [1, Chapter 2.9, Theorem 3], we conclude that these points are also *Liapunov stable*, but not *asymptotically stable*.

μ	$\mathbf{x}^*$	Class	Stability
$\mu < 0$	$(0,0)$	Center	Liapunov Stable
$\mu > 0$	$(0,0)$	Saddle	Unstable
$\mu > 0$	$(\sqrt{\mu}, 0)$	Center	Liapunov Stable
$\mu > 0$	$(-\sqrt{\mu}, 0)$	Center	Liapunov Stable

Table 1: Classification and stability summary of Section 1.b

1.c) Sketching the Phase Portrait & Flow

Using the results from Section 1.b, we can sketch the phase portraits for both cases $\mu > 0$ and $\mu < 0$ as seen in Figure 1.

The homoclinic orbit shown in purple can be observed encircling the stable centres at $(\pm\sqrt{\mu}, 0)$ and crossing at the saddle point in $(0,0)$.

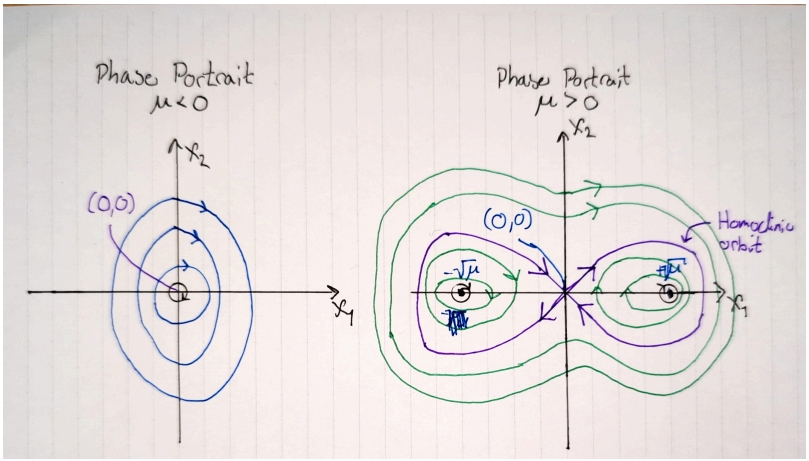


Figure 1: Sketch of the phase plane of vector field $\mathbf{f}$ from Equation 2 for $\mu > 0$ (left) and $\mu < 0$ (right).

We additionally plot the numerically computed phase portraits rendered using Python's `matplotlib` library as shown in Figure 2, which confirms our qualitative sketches from Figure 1.

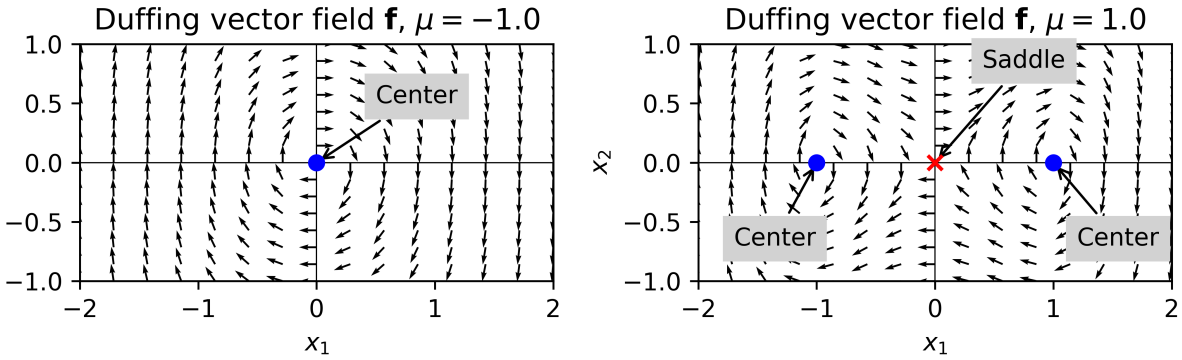


Figure 2: Sketch of the phase portrait of vector field $\mathbf{f}$ from Equation 2 for $\mu > 0$ (left) and $\mu < 0$ (right). Critical points are marked in red (saddles) and blue (centers). Arrows indicate the flow of the system.

1.d) Analytical Homoclinic Orbit (1st Quadrant)

Before we begin discussions, it helps to formulate a more intuitive definition of *homoclinic orbits* [1, Chapter 3.3]. A homoclinic orbit is a trajectory in the phase space of a dynamical system that starts and ends at the same saddle point. In other words, it is a path that leaves a saddle point along its unstable manifold and then returns to the same saddle point along its stable manifold.

In the phase portrait for $\mu > 0$ shown on the right panel of Figure 2, we can observe a homoclinic orbit associated with the saddle point x_0^* at the origin $(0, 0)$. These orbits leave the saddle point along its unstable manifold and return to it along its stable manifold, forming closed loops around the other critical points $x_{1,2}^*$ at $(\pm\sqrt{\mu}, 0)$. We note that the purple cycle shown in the right panel of Figure 1 is determined by the equilibrium point at the origin, which in combination with the homoclinic orbit forms a *separatrix cycle* [1, Chapter 3.3]. This cycle separates different types of motion in the phase space, distinguishing between trajectories that remain close to the center points and those that diverge away.

Now we can determine a function $h \in C^1$, that prescribes these homoclinic orbits in the phase space, where notably the treatment of h may require piecewise definitions to avoid construction of a many-valued function. This function $h : \mathbb{R} \to \mathbb{R}$ will describe a graph in the phase plane according to

$$x_2 = h(x_1) \tag{23}$$

We require that a solution trajectory lying on the curve formed by $h(x_1)$ must remain on that curve for all time t, which requires that the derivatives of $h(x_1)$ and x_2 are equal along that curve, yielding the following ordinary differential equation involving h:

$$\dot{x}_2 = \frac{\mathrm{d}h(x_1)}{\mathrm{d}t} = \frac{\mathrm{d}h}{\mathrm{d}x_1}\dot{x}_1 \Leftrightarrow \frac{\mathrm{d}h}{\mathrm{d}x_1} = \frac{\dot{x}_2}{\dot{x}_1} = \frac{\mu x_1 - x_1^3}{x_2} \tag{24}$$

Which we may solve by separation of variables:

$$\begin{aligned}
\frac{\mathrm{d}h}{\mathrm{d}x_1} &= \frac{\mu x_1 - x_1^3}{h(x_1)} \\
&\Updownarrow \\
h(x_1)\,\mathrm{d}h &= (\mu x_1 - x_1^3)\,\mathrm{d}x_1 \\
&\Updownarrow \\
\int h(x_1)\,\mathrm{d}h &= \int (\mu x_1 - x_1^3)\,\mathrm{d}x_1 \\
&\Updownarrow \\
\frac{1}{2}h(x_1)^2 &= \frac{\mu}{2}x_1^2 - \frac{1}{4}x_1^4 + C \\
&\Updownarrow \\
h(x_1)^2 &= \mu x_1^2 - \frac{1}{2}x_1^4 + 2C \\
&\Updownarrow \\
h(x_1) &= \cancel{\pm}\sqrt{\mu x_1^2 - \frac{1}{2}x_1^4 + 2C}
\end{aligned} \tag{25}$$

Where $C \in \mathbb{R}$ is the integration constant and the negative root is discarded on the grounds that $x_2 > 0$ if we restrict ourselves to the first quadrant of the phase plane.

Since we require that the homoclinic orbit passes through the origin, we impose $h(0) = 0 = \sqrt{0 + 0 + 2C} = 0$, thus yielding the particular solution:

$$h(x_1) = x_1\sqrt{\mu - \frac{1}{2}x_1^2} \qquad x_1 \in \left[0, \sqrt{2\mu}\right] \tag{26}$$

Where we have found the domain of h by requiring that the radicand is non-negative to obtain the interval:

$$\mu - \frac{1}{2}x_1^2 \geq 0 \Leftrightarrow x_1^2 \leq 2\mu \Leftrightarrow x_1 \in \left[0, \sqrt{2\mu}\right] \tag{27}$$

Regarding the map $h(x_1)$ when considered only in the first quadrant of the phase plane, ($x_1 > 0, x_2 > 0$), we note that h is *not* a one-to-one mapping between x_1 and x_2, though it is non-negative in this quadrant and as such the we find $0 < x_2 = \dot{x}_1$, which implies that the position along x_1 is monotonically increasing with time t, which could be what the question hints at. As such, we conclude that both x_1 and (x_1, x_2) are one-to-one mappings with respect to time t in this quadrant.

To prove that $h(x_1)$ cannot be one-to-one, we may simply refer to the graph in Figure 2 and pick a counter example of two points in the first quadrant for which the second coordinate x_2 is the same. Alternatively, the derivative of $h(x_1)$ may be investigated to find that both negative and positive slopes exist in the first quadrant, indicating that h is not monotonic and thus not one-to-one.

Alternatively, the assignment may have meant to say that $h(x_1)$ is single-valued in the first quadrant, which is true as we have taken the positive root of the homoclinic orbit function.

Note that we can interpret the maximum possible value for $x_1 = \sqrt{2\mu}$ as the turning point of the homoclinic orbit in the first quadrant, after which it returns to the origin along the stable manifold along the orbit on the fourth quadrant ($x_1 > 0, x_2 < 0$). Thus it starts at the origin $h(x_1 = 0) = 0$ and transitions into the fourth quadrant at $h(x_1 = \sqrt{2\mu}) = 0$, see Figure 2.

Lastly we note that while the homoclinic orbit is closed, it takes infinite time to traverse it due to the flow slowing down as it approaches the saddle point at the origin. This is a common characteristic of homoclinic orbits in dynamical systems.

1.e) Plotting the Homoclinic Orbit (1st quadrant)

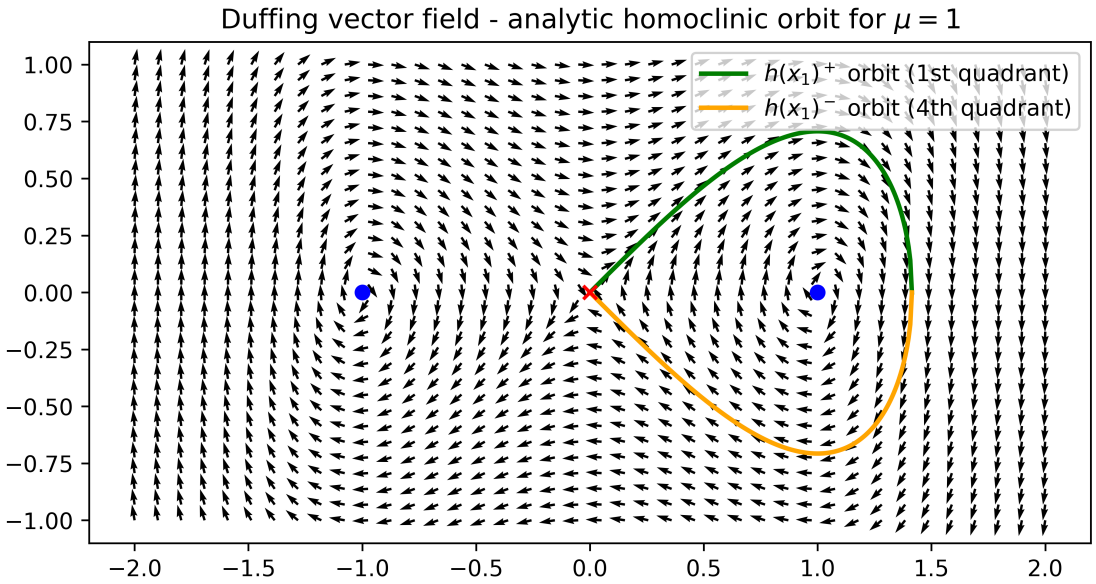


Figure 3: Plot of the homoclinic orbit $x_2 = h(x_1)$ from Equation 24 for $\mu > 0$. The orbit starts and ends at the saddle point at the origin $(0, 0)$, forming a closed loop around the other critical points.

In Figure 3 we observe that alongside the homoclinic orbital path in the fourth quadrant, the orbit path in the first quadrant encircles a stable center at $(\sqrt{\mu}, 0)$ with each trajectory within the encircled area corresponding to a different energy level of the system, or equivalently a unique bounded orbit around the center. Each of these trajectories are invariant sets of the system. This aligns with our intuition of a conserved energy system and is consistent with Figure 1 and Figure 2.

We note that all trajectories on the orbit are tangent to to it as expected and required by the invariance condition in Equation 24.

Lastly, we again note that $h(x_1)$ forms a boundary between different types of motion in the phase space, where trajectories below the epigraph correspond to bounded orbits around the center at $(\sqrt{\mu}, 0)$, while those above it form cycles around both centers.

1.f) Analytical Homoclinic Orbit (4th Quadrant)

Given the nature of Equation 26, for the fourth quadrant of the phase plot ($x_1 > 0, x_2 < 0$), it is merely taking the negative root of the homoclinic orbit function, i.e.:

$$h(x_1)^- = -x_1\sqrt{\mu - \frac{x_1^2}{2}} \qquad \forall x_1 \in \left[0, \sqrt{2\mu}\right] \tag{28}$$

The resulting graph of $h(x_1)$ can be seen in Figure 3 in orange, where it mirrors the first quadrant homoclinic orbit across the x_1-axis. All statements concerning turning points and behaviour apply analogously.

2) Problem 2 — Bifurcation

A bifurcation parameter is a parameter in a dynamical system that, when varied, causes a change in the system's behavior or structure. Bifurcations often lead to the emergence or disappearance of critical points, periodic orbits, or changes in stability at those points.

2.a) Bifurcation parameter & diagram

As our analysis in Section 1 showed, the parameter μ in the Duffing equation from Equation 2 acts as a bifurcation parameter. Specifically, we observed an emergence of two additional critical points $x^*_{1,2}$ at $(\pm\sqrt{\mu}, 0)$ when μ transitions from negative to positive values. This indicates a bifurcation occurring at the critical value $\mu = 0$.

With this, we are able to sketch a bifurcation diagram illustrating how the critical points of the system change as the bifurcation parameter μ varies as shown in Figure 4.

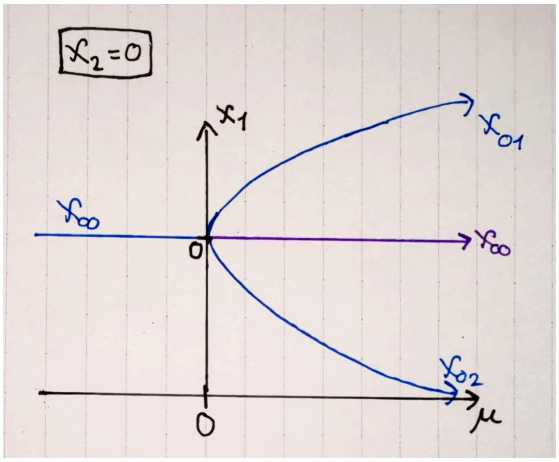


Figure 4: Sketch of the bifurcation diagram illustrating the emergence of critical points in the Duffing equation from Equation 2 as the bifurcation parameter μ varies. Blue lines indicate centers, whereas the purple line indicates a saddle point.

Additionally, we plot the numerically computed bifurcation diagram rendered using Python's `matplotlib` library as shown in Figure 5, which confirms our qualitative sketch from Figure 4.

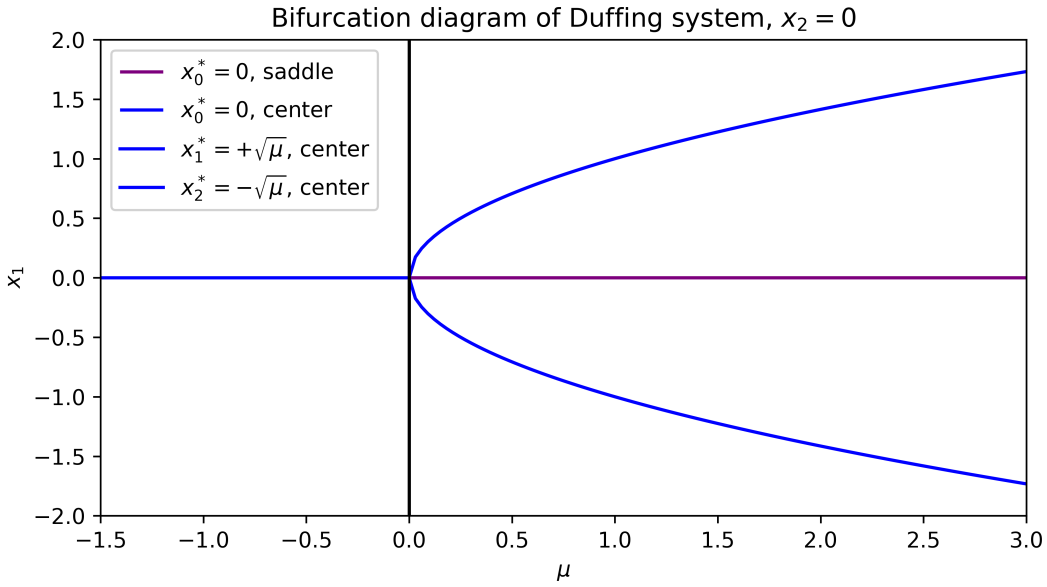


Figure 5: Bifurcation diagram illustrating the emergence of critical points in the Duffing equation from Equation 2 as the bifurcation parameter μ varies.

We observe that the critical point at the origin $(0, 0)$ transitions from a stable center (blue line) for $\mu < 0$ to an unstable saddle point (purple line) for $\mu > 0$. Simultaneously, two new stable centers

(blue lines) emerge symmetrically at $(\pm\sqrt{\mu}, 0)$ for $\mu > 0$. This bifurcation behavior is characteristic of a pitchfork bifurcation.

2.b) Type of Bifurcation

The bifurcation observed in Figure 5 from Duffing equation from Equation 2 as the parameter μ crosses zero is an archetypical example of a **pitchfork bifurcation** [1, Chapter 4.2, Figure 6], sometimes referred to as a **supercritical pitchfork bifurcation** in other literature, where supercriticality indicates that as the bifurcation parameter μ passes through a transition value, here $\mu = 0$, a single center stable critical point at the origin $(0, 0)$ for $\mu < 0$ becomes an unstable saddle point for $\mu > 0$, and two new stable critical points (at $(\pm\sqrt{\mu}, 0)$) emerge symmetrically on either side of the original point.

2.c) Hyperbolicity for bifurcation transition $\mu = 0$

We determine the hyperbolicity of the system at $\mu = 0$ by computing the Jacobian:

$$Df(x_0^*) = Df(0,0) = \begin{bmatrix} 0 & 1 \\ -3x_1^2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \tag{29}$$

Of which we may determine the eigenvalues by solving the characteristic polynomial:

$$p_{x_0^*}(\lambda) = \det(Df(x_0^*) - \lambda I) = \lambda^2 = 0 \Rightarrow \lambda_{1,2} = 0 \tag{30}$$

Clearly, this is non-hyperbolic, as hyperbolicity requires that all eigenvalues have non-zero real parts.

It follows that the critical point is non-hyperbolic at the bifurcation point $\mu = 0$. Hence, the Hartman-Grobman Theorem [1, Chapter 2.8] does not apply here, and we cannot draw conclusions about the stability of the critical point based solely on the linearized system.

2.d) Lyapunov Stability for non-hyperbolic case

As we determined in Section 2.c, the critical point x_0^* at the bifurcation point $\mu = 0$ is non-hyperbolic and we cannot conclude stability from the linearized system alone. Hence, we can again apply the theorems of [1, Chapter 2.10] as done for the non-hyperbolic cases in Section 1.b.

We recall the chosen Liapunov function from Equation 16:

$$V(x_1, x_2) = \frac{1}{2}x_2^2 - \frac{1}{2}\cancel{\mu}\cancel{x_1^2} + \frac{1}{4}x_1^4 \tag{31}$$

Next, we compute the time derivative of V along the trajectories / solutions $x_1(t), x_2(t)$ of the system $\boldsymbol{f}$ for $\mu = 0$:

$$\begin{aligned} \dot{V}(x_1(t), x_2(t)) &= \nabla_t V(x_1(t), x_2(t))(\dot{x}_1(t), \dot{x}_2(t)) && \text{via chain rule} \\ &= \frac{\partial V}{\partial x_1}\dot{x}_1 + \frac{\partial V}{\partial x_2}\dot{x}_2 && \text{inserting into function} \\ &= x_1^3\dot{x}_1 + x_2\dot{x}_2 \\ &= x_1^3 x_2 + x_2(\mu - x_1^3) \\ &= x_1^3 x_2 + x_2(-x_1^3) \\ &= 0 \end{aligned} \tag{32}$$

Since the requirements of [1, Chapter 2.9, Theorem 3] given in Equation 13 are satisfied, we may again leverage this theorem alongside the conclusion that $\dot{V} = 0$ to determine the stability of the critical point at the origin for $\mu = 0$ as *stable* or *Lyapunov stable*, but not *asymptotically stable*, which would require $\dot{V} < 0$.

As such, the origin is a stable center (Lyapunov stable), but not asymptotically stable.

2.e) Contour Plot of Lyapunov Function

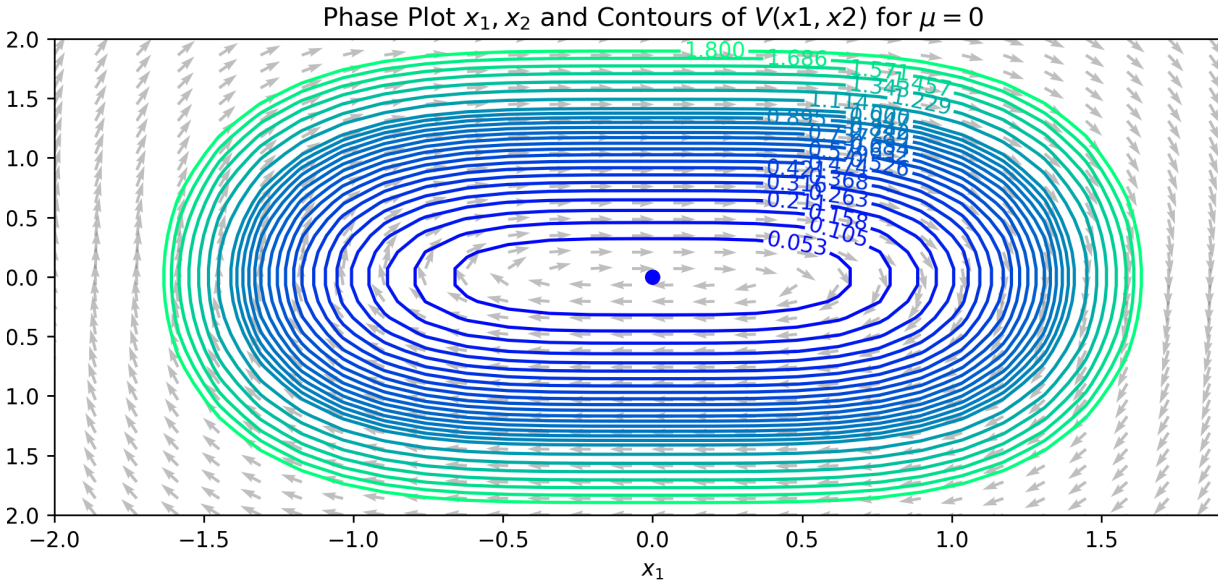


Figure 6: Contour plot of the Lyapunov function $V(x_1, x_2)$ from Equation 31 for $\mu = 0$. The contours represent level sets of constant energy in the system. The plot is overlaid with vector field arrows.

As clearly visible in Figure 6, the contours of the Lyapunov function $V(x_1, x_2)$ from Equation 31 represent not only level sets of constant energy but also trajectories as solutions of the system Equation 2 for $\mu = 0$. This is consistent with our earlier analysis in Section 2.d, where we found that the time derivative of the Lyapunov function $\dot{V}$ is identically zero along trajectories, indicating that energy is conserved.

Further we observe, that the level sets along with trajectories of the vector field form elliptical closed orbits around the origin. The elliptical shape could be attributed to the stretch of the system along the x_1-axis, due to the quartic term in V (Equation 31), compared to only the quadratic term for x_2.

2.f) Stability for $(0, 0)$ at $\mu = 0$

The required theorem from [1, Chapter 2.11, p.151, Theorem 2] states:

For when matrix A of a linearized system has two zero eigenvalues, i.e. $\det(A) = \mathrm{tr}(A) = 0$, but $A \neq 0$, the system can be put into a 'normal' form:

$$\begin{aligned} \dot{x} &= y \\ \dot{y} &= a_k x^k [1 + h(x)] + b_n x^n y [1 + g(x)] + y^2 R(x, y) \end{aligned} \tag{33}$$

where $h(x), g(x), R(x, y)$ are analytic in the neighborhood of the origin with $h(0) = g(0) = 0$ and $k \geq 2, n \geq 1$.

2.f.1) Perko Theorem [1, Chapter 2.11, P. 151, Theorem 2]

Let $k = 2m + 1$ with $m \geq 1$ in Equation 33 and let $\lambda = b_n^2 + 4(m + 1)a_k$. Then if $a_k > 0$, the origin is a topological saddle. If $a_k < 0$, the origin is
(1) a focus or a center if $b_n = 0$ and also if $b_n \neq 0$ and $n > m$ or if $n = m$ and $\lambda < 0$,
(2) a node if $b_n \neq 0$, n is an even number and $n < m$ and also if $b_n \neq 0$, n is an even number, $n = m$ and $\lambda \geq 0$ and
(3) a critical point with an elliptic domain if $b_n \neq 0$, n is an odd number and $n < m$ and also if $b_n \neq 0$, n is an odd number, $n = m$ and $\lambda \geq 0$.

From this theorem, we can conclude that our system Equation 2 at the critical point $(0,0)$ for $\mu = 0$ can be transformed into the normal form Equation 33 with:

$$\begin{aligned} k &= 3 \\ a_k &= -1 \\ b_n &= 0 \\ h(x) &= 0 \\ R(x,y) &= 0 \end{aligned} \tag{34}$$

Thus, we have a focus or a center according to case (1) in the theorem, which is consistent with our earlier findings in Section 2.d using the Lyapunov function, where we concluded that the origin is a stable center (Lyapunov stable) but not asymptotically stable. The theorem is notably *not* able to distinguish between a focus and a center in this case.

Bibliography

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